

Metric structures and signatures

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1 Relevant basic facts about metric spaces

Let (M, d) be a metric space. We say this space is bounded if there is a real number B such that $d(x, y) \leq B$ for all $x, y \in M$. The diameter of (M, d) is the smallest such number B . We say this space is complete if every Cauchy sequence of points in M has a limit that is also in M .

Suppose (M_i, d_i) are metric spaces for $i = 1, \dots, n$ and we take M to be the product $M = M_1 \times \dots \times M_n$. In this seminar we will always regard M as being equipped with the maximum metric, defined for $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n)$ by

$$d(x, y) = \max\{d_i(x_i, y_i) : i = 1, \dots, n\}.$$

We will now assume that every metric space is bounded and complete.

1.1 Definition: modulus of uniform continuity

A *modulus of uniform continuity* is any function

$$\Delta : (0, 1] \rightarrow (0, 1].$$

If (M, d) and (M', d') are metric spaces and $f : M \rightarrow M'$ is any function, we say that $\Delta : (0, 1] \rightarrow (0, 1]$ is a *modulus of uniform continuity for f* if for every $\varepsilon \in (0, 1]$ and every $x, y \in M$ we have

$$d(x, y) < \Delta(\varepsilon) \Rightarrow d'(f(x), f(y)) \leq \varepsilon. \quad (\text{UC})$$

We say that f is *uniformly continuous* if it has a modulus of uniform continuity.

Note that this definition is equivalent to the definition of uniform continuity, which we got to know in Calculus 2. To prove that, we define $\Delta(\varepsilon) := \delta$ and adjust $d'(f(x), f(y)) < \varepsilon$

1.2 Lemma: Composition of moduli of uniform continuity

Suppose $f : M \rightarrow M'$ and $f' : M' \rightarrow M''$ are functions between metric spaces M, M', M'' . Suppose Δ is a modulus of uniform continuity for f and Δ' is a modulus of uniform continuity for f' .

Then the composition $f' \circ f$ is uniformly continuous; indeed, for each $r \in (0, 1)$ the function $\Delta(r \Delta'(\varepsilon))$ is a modulus of uniform continuity for $f' \circ f$.

Proof. Consider $\varepsilon \in (0, 1]$ and $x, y \in M$. Δ and Δ' are moduli of uniform continuity. Thus, we have that if

$$d(x, y) < \Delta(r \cdot \Delta'(\varepsilon)) \Rightarrow d'(f(x), f(y)) \leq r \cdot \Delta'(\varepsilon) < \Delta'(\varepsilon),$$

because $r \in (0, 1)$. And since

$$d'(f(x), f(y)) < \Delta'(\varepsilon),$$

we obtain

$$d''(f'(f(x)), f'(f(y))) \leq \varepsilon.$$

□

Let M, M' be metric spaces (with metrics d and d' , respectively) and let f and $(f_n : n \geq 1)$ be functions from M into M' . Recall that $(f_n : n \geq 1)$ converges uniformly to f on M if

$$\forall \varepsilon > 0 \exists N \forall n > N \forall x \in M \quad d'(f_n(x), f(x)) \leq \varepsilon.$$

1.3 Lemma: modulus of uniform continuity and uniformly convergence

Let M, M', f and $(f_n : n \geq 1)$ be as above, and suppose $(f_n : n \geq 1)$ converges uniformly to f on M . If each of the functions $f_n : M \rightarrow M'$ is uniformly continuous, then f must also be uniformly continuous.

Indeed, a modulus of uniform continuity for f can be obtained from moduli Δ_n for f_n (for each $n \geq 1$) and from a function $N : (0, 1] \rightarrow \mathbb{N}$ that satisfies

$$\forall \varepsilon > 0 \ \forall n > N(\varepsilon) \ \forall x \in M, \quad d'(f_n(x), f(x)) \leq \varepsilon.$$

Proof. Note that there is always such a function N , as stated in 1.3., because of the definition of uniform convergence

Let $x, y \in M$. We define

$$\Delta(\varepsilon) := \Delta_n(\varepsilon/3).$$

for $n = N(\varepsilon/3) + 1$. Because each Δ_n is a modulus of uniform continuity for f_n , we know that

$$d(x, y) < \Delta_n(\varepsilon/3) \quad \Rightarrow \quad d'(f_n(x), f_n(y)) \leq \varepsilon/3,$$

Fix $\varepsilon \in (0, 1]$. If $d(x, y) < \Delta(\varepsilon)$;

$$\begin{aligned} d'(f(x), f(y)) &\leq d'(f(x), f_n(x)) + d'(f_n(x), f_n(y)) + d'(f_n(y), f(y)) \\ &\leq \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon. \end{aligned}$$

□

The first and third term in line 1 is at most $\varepsilon/3$, because of the definition of uniform convergence and the second one is at most $\varepsilon/3$, because

$$\Delta(\varepsilon) := \Delta_n(\varepsilon/3).$$

1.4 Definition: $\sup_y f$ and $\inf_y f$

Suppose M, M' are metric spaces and $f : M \times M' \rightarrow \mathbb{R}$ is a bounded function. We define new functions $\sup_y f$ and $\inf_y f$ from M to $\mathbb{R}$ by

$$(\sup_y f)(x) := \sup\{f(x, y) : y \in M'\}, \quad (\inf_y f)(x) := \inf\{f(x, y) : y \in M'\},$$

for all $x \in M$.

Note that these new functions map M into the same closed bounded interval in $\mathbb{R}$ that contained the range of f .

Our perspective is that $\sup_y$ and $\inf_y$ are quantifiers that bind or eliminate the variable y , analogous to the way $\forall$ and $\exists$ are used in ordinary first-order logic.

1.5 Proposition: Properties of $\sup_y f$ and $\inf_y f$

Suppose M, M' are metric spaces and f is a bounded, uniformly continuous function from $M \times M'$ to $\mathbb{R}$. Let Δ be a modulus of uniform continuity for f .

Then $\sup_y f$ and $\inf_y f$ are bounded, uniformly continuous functions from M to $\mathbb{R}$, and Δ is a modulus of uniform continuity for both of them.

Proof. Only for $\sup_y f$, it is analogous for $\inf_y f$. $\sup_y f$ is obviously bounded, when f is bounded. Fix $\varepsilon > 0$ and consider $u, v \in M$ so that $d(u, v) < \Delta(\varepsilon)$. Then for every $z \in M'$ we have the following inequalities:

$$1. \quad f(v, z) \leq f(u, z) + \varepsilon.$$

$$\text{Proof. } d((v, z), (u, z)) < \Delta(\varepsilon) \Rightarrow d'(f((v, z)), f((u, z))) \leq \varepsilon \Leftrightarrow |f((v, z)) - f((u, z))| \leq \varepsilon$$

$$\text{It is known that } f((v, z)) - f((u, z)) \leq |f((v, z)) - f((u, z))|$$

$$\Rightarrow f(v, z) \leq f(u, z) + \varepsilon.$$

$$2. \quad f(u, z) + \varepsilon \leq (\sup_y f)(u) + \varepsilon.$$

Thus, we get

$$f(v, z) \leq (\sup_y f)(u) + \varepsilon.$$

Choose $z \in M'$ such that $f(u, z) = (\sup_y f)(u)$. Then

$$\begin{aligned} |(\sup_y f)(u) - (\sup_y f)(v)| &= |(\sup_y f)(u) - f(u, z) + f(u, z) - (\sup_y f)(v)| \\ &= |((\sup_y f)(u) - f(u, z)) + (f(u, z) - (\sup_y f)(v))| \\ &\leq |0 + \varepsilon| = \varepsilon. \end{aligned}$$

So $\sup_y f$ has a modulus of uniform continuity, which means that it is uniformly continuous. $\square$

1.6 Proposition: continuous function of arbitrary functions

Let $F, G : X \rightarrow [0, 1]$ be arbitrary functions such that

$$(*) \quad \forall \varepsilon > 0 \exists \delta > 0 \forall x \in X (F(x) \leq \delta \Rightarrow G(x) \leq \varepsilon).$$

Then there exists an increasing, continuous function $\alpha : [0, 1] \rightarrow [0, 1]$ such that $\alpha(0) = 0$ and

$$(**) \quad \forall x \in X, \quad G(x) \leq \alpha(F(x)).$$

Proof. Define a (possibly discontinuous) function $g : [0, 1] \rightarrow [0, 1]$ by

$$g(t) = \sup\{G(x) : F(x) \leq t\}, \quad t \in [0, 1].$$

- **g is increasing.** This is a direct consequence of the definition of a supremum.
- $G(x) \leq g(F(x))$. *Proof.* Assume that there exists $x' \in X$ such that $G(x') > g(F(x'))$. This means that

$$G(x') > g(F(x')) = \sup\{G(x) : F(x) \leq F(x')\} \geq G(x'),$$

because $F(x') \leq F(x')$ which is a contradiction.

- $g(0) = 0$. *Proof.* We only need to check that $g(0) > 0$ is impossible. Assume that

$$g(0) = \sup\{G(x) : F(x) \leq 0\} = G(x_0) =: \varepsilon_0 > 0.$$

Fix $\varepsilon := \varepsilon_0/2$. According to $(*)$, we find $\delta > 0$ such that $F(x_0) \leq \delta \Rightarrow G(x_0) \leq \varepsilon$. That means $G(x_0) \leq \varepsilon_0/2 < \varepsilon_0 = G(x_0)$, which is a contradiction.

- $\lim_{t \rightarrow 0} g(t) = 0$. *Proof.* Choose $\varepsilon > 0$. According to $(*)$, there exists $\delta > 0$ such that $F(x) \leq \delta \Rightarrow G(x) \leq \varepsilon$. Let $0 < t \leq \delta$. Because $t \leq \delta$ and g is increasing, we get

$$g(t) = \sup\{G(x) : F(x) \leq t\} \leq \sup\{G(x) : F(x) \leq \delta\} \leq \varepsilon.$$

Thus, $g(t) \leq \varepsilon$. Hence $\forall \varepsilon > 0 \exists \delta > 0 \forall t \in (0, \delta] : g(t) \leq \varepsilon$, which is the definition of $\lim_{t \rightarrow 0} g(t) = 0$. *Remark:* We can assume that δ (and thus t) ≤ 1 so that we stay within bounds.

- **Properties of α .** *Proof.* To complete the proof we construct an increasing, continuous function $\alpha : [0, 1] \rightarrow [0, 1]$ such that $\alpha(0) = 0$ and $g(t) \leq \alpha(t)$ for all $t \in [0, 1]$.

Let $(t_n : n \in \mathbb{N})$ be any decreasing sequence in $[0, 1]$ with $t_0 = 1$ and $\lim_{n \rightarrow \infty} t_n = 0$. Define $\alpha : [0, 1] \rightarrow [0, 1]$ by setting $\alpha(0) = 0$, $\alpha(1) = 1$, and $\alpha(t_n) = g(t_{n-1})$ for all $n \geq 1$, and by taking α to be linear on each interval $[t_{n+1}, t_n]$, $n \in \mathbb{N}$.

The direct consequence of those definitions is that

$$g(t_n) = \alpha(t_{n+1}) \leq \alpha(t_n),$$

because α is linear. Let $x \in (t_{n+1}, t_n)$, so $x = t_{n+1} + \varepsilon$ with $\varepsilon > 0$. Because α is linear, we get

$$\alpha(x) = mx + b.$$

with

$$m = \frac{\alpha(t_n) - \alpha(t_{n+1})}{t_n - t_{n+1}} = \frac{g(t_{n-1}) - g(t_n)}{t_n - t_{n+1}}, \quad b = \alpha(t_n) - mt_n = g(t_{n-1}) - mt_n.$$

And thus, we get

$$\alpha(x) = \frac{g(t_{n-1}) - g(t_n)}{t_n - t_{n+1}}x + g(t_{n-1}) - \frac{g(t_{n-1}) - g(t_n)}{t_n - t_{n+1}}t_n = \frac{g(t_{n-1}) - g(t_n)}{t_n - t_{n+1}}(x - t_n) + g(t_{n-1}).$$

Because g is increasing, we get the inequality

$$g(x) = g(t_{n+1} + \varepsilon) \leq g(t_n).$$

We obtain

$$\alpha(x) - g(t_n) = \frac{g(t_{n-1}) - g(t_n)}{t_n - t_{n+1}}(x - t_n) + g(t_{n-1}) - g(t_n) = \left(\frac{x - t_n}{t_n - t_{n+1}} + 1 \right) (g(t_{n-1}) - g(t_n)).$$

Now we analyze both terms:

$$g(t_{n-1}) - g(t_n) \geq 0, \quad \text{since } g \text{ is increasing and } t_{n-1} > t_n,$$

and

$$\frac{x - t_n}{t_n - t_{n+1}} + 1 = \frac{x - t_n + t_n - t_{n+1}}{t_n - t_{n+1}} = \frac{\varepsilon}{t_n - t_{n+1}} > 0.$$

Thus $\alpha(x) - g(t_n) \geq 0$. To sum up, we get

$$g(x) \leq g(t_n) \leq \alpha(x) \quad \text{for } x \in (t_{n+1}, t_n).$$

Together with $g(0) = \alpha(0) = 0$, this shows that $g(t) \leq \alpha(t)$ for all $t \in [0, 1]$.

□

2 Metric Structures and signatures

Let (M, d) be a complete, bounded metric space.

2.1 Definition: metric structures

A *predicate* on M is a uniformly continuous function from M^n (for some $n \geq 1$) into some bounded interval in $\mathbb{R}$. A *function* or *operation* on M is a uniformly continuous function from M^n (for some $n \geq 1$) into M . In each case n is called the *arity* of the predicate or function.

A *metric structure* $\mathcal{M}$ based on (M, d) consists of a family $(R_i : i \in I)$ of predicates on M , a family $(F_j : j \in J)$ of functions on M , and a family $(a_k : k \in K)$ of distinguished elements of M (we can view them as constants like in first-order logic).

When we introduce such a metric structure, we will often denote it as

$$\mathcal{M} = (M, R_i, F_j, a_k : i \in I, j \in J, k \in K).$$

Any of the index sets I, J, K is allowed to be empty. Indeed, they might all be empty, in which case $\mathcal{M}$ is a pure bounded metric space.

2.2 Examples: metric structures

We give a number of examples of metric structures.

1. A complete, bounded metric space (M, d) with no additional structure. Note that d is a predicate with arity 2. Thus (M, d) has to be bounded, because then we know, that d maps into a bounded Interval in $\mathbb{R}$. Our sets of functions and constants would be empty in this case.

2. A structure $\mathcal{M}$ in the usual sense from first-order logic. Consider $\mathcal{A}$ to be an S-structure. One puts the discrete metric on the underlying set A ($d(a, b) = 1$ when a, b are distinct), so that d becomes a predicate like in the first example. We can just "copy" our functions from $\mathcal{A}$ into $\mathcal{M}$ and our constants. For each relation in $\mathcal{A}$, we have to create a predicate function. So we can consider a relation as a predicate taking values ("truth values") in the set $\{0, 1\}$.

Note that defining the discrete metric on our underlying set A does not expand the sense of $\mathcal{A}$, because we defined "=" as being a logical symbol and therefore it is always included in first-order logic.

3. The unit ball B of a Banach space (a complete, normed space) X over $\mathbb{R}$ or $\mathbb{C}$: as functions we may take the maps $f_{\alpha, \beta}$, defined by $f_{\alpha, \beta}(x, y) = \alpha x + \beta y$, for each pair of scalars satisfying $|\alpha| + |\beta| \leq 1$ (these are uniformly continuous, so we obtain functions of arity 2); the norm may be included as a predicate (because our underlying set is the unit ball, we know that it maps into a bounded interval, so it is a predicate with arity 1), and we may include the additive identity 0_X (0-vector) as a distinguished element.

2.3 Definition: signatures

To each metric structure $\mathcal{M}$ we associate a *signature* L as follows. To each predicate R of $\mathcal{M}$ we associate a predicate symbol P and an integer $a(P)$ which is the arity of R ; we denote R by $P^{\mathcal{M}}$. To each function F of $\mathcal{M}$ we associate a function symbol f and an integer $a(f)$ which is the arity of F ; we denote F by $f^{\mathcal{M}}$. Finally, to each distinguished element a of $\mathcal{M}$ we associate a constant symbol c ; we denote a by $c^{\mathcal{M}}$.

In addition, a signature for metric structures must specify more: for each predicate symbol P , it must provide a closed bounded interval I_P of real numbers and a modulus of uniform continuity Δ_P . These should satisfy the requirements that $P^{\mathcal{M}}$ takes its values in I_P and that Δ_P is a modulus of uniform continuity for $P^{\mathcal{M}}$.

In addition, for each function symbol f , L must provide a modulus of uniform continuity Δ_f , and this must satisfy the requirement that Δ_f is a modulus of uniform continuity for $f^{\mathcal{M}}$.

Finally, L must provide a non-negative real number D_L which is a bound on the diameter of the complete metric space (M, d) on which $\mathcal{M}$ is based.

We sometimes denote the metric d given by $\mathcal{M}$ as $d^{\mathcal{M}}$. However, we also find it convenient often to use the same notation d for both the logical symbol representing the metric and its interpretation in $\mathcal{M}$.

2.4 Definition: L -structures

When these requirements are all met and when the predicate, function, and constant symbols of L correspond exactly to the predicates, functions, and distinguished elements of which $\mathcal{M}$ consists, then we say that $\mathcal{M}$ is an L -structure.

The key added features of a signature L in the metric structure setting are that L specifies

1. a bound on the diameter of the underlying metric space,
2. a modulus of uniform continuity for each predicate and function, and
3. a closed bounded interval of possible values for each predicate.

For simplicity, and without losing any generality, we will usually assume that our signatures L satisfy $D_L = 1$ and $I_P = [0, 1]$ for every predicate symbol P .

2.5 Definition: embeddings

Let L be a signature for metric structures and suppose $\mathcal{M}$ and $\mathcal{N}$ are L -structures.

An *embedding* from $\mathcal{M}$ into $\mathcal{N}$ is a metric space isometry ($d^{\mathcal{M}}(x, y) = d^{\mathcal{N}}(T(x), T(y))$) for all $x, y \in M$)

$$T : (M, d^{\mathcal{M}}) \rightarrow (N, d^{\mathcal{N}})$$

that commutes with the interpretations of the function and predicate symbols of L in the following sense:

- Whenever f is an n -ary function symbol of L and $a_1, \dots, a_n \in M$, we have

$$f^{\mathcal{N}}(T(a_1), \dots, T(a_n)) = T(f^{\mathcal{M}}(a_1, \dots, a_n)).$$

- Whenever c is a constant symbol of L , we have

$$c^{\mathcal{N}} = T(c^{\mathcal{M}}).$$

- Whenever P is an n -ary predicate symbol of L and $a_1, \dots, a_n \in M$, we have

$$P^{\mathcal{N}}(T(a_1), \dots, T(a_n)) = P^{\mathcal{M}}(a_1, \dots, a_n).$$

Note that all embeddings are injective per definition.

2.6 Definition: Isomorphism

An *isomorphism* is a surjective embedding. We say that $\mathcal{M}$ and $\mathcal{N}$ are *isomorphic*, and write $\mathcal{M} \cong \mathcal{N}$, if there exists an isomorphism between them. (Sometimes we say *isometric isomorphism* to emphasize that isomorphisms must be distance preserving.)

An *automorphism* of $\mathcal{M}$ is an isomorphism between $\mathcal{M}$ and itself.

We say that $\mathcal{M}$ is a *substructure* of $\mathcal{N}$ (and write $\mathcal{M} \subseteq \mathcal{N}$) if $M \subseteq N$ and the inclusion map from M into N is an embedding of $\mathcal{M}$ into $\mathcal{N}$.

References

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